

Tianyou Lai

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Education

Carnegie Mellon University M.S. in Artificial Intelligence Systems Management	Aug. 2026 – Dec. 2027 (expected)
Lanzhou University B.S. in Data Science & Big Data Technology GPA: 87.34/100	Sep. 2021 – Jul. 2025
The University of Texas at Austin Software Engineering Exchange Program GPA: 4.0/4.0	Jan. 2024 – Feb. 2024

Technical Skills

Trading and modeling: Python, SQL, C/C++; CTP-API, Simnow, TWAP, VWAP, TA-Lib; time-series forecasting and model evaluation.
AI and data systems: PyTorch, TensorFlow, federated learning, LLM evaluation; MySQL, Hadoop, Spark, Linux.
Analysis and presentation: Matplotlib, Seaborn, Excel, Tableau, PowerPoint.

Professional Experience

Guotai Junan Futures Quantitative Research Assistant • Extended a Python futures trading system with position and batch-order functions; debugged C++/Python CTP-API integrations in Simnow and implemented TWAP/VWAP execution with TA-Lib indicators.	Mar. – Apr. 2024
Sichuan DeepGraph Intelligent Technology Data Science Intern • Validated datasets in Python, queried graph relationships with Ultpa, and supported analysis-algorithm optimization and OpenStack deployment.	Sep. 2024 – Feb. 2025
Lanzhou Bronze Ding Intelligent Technology Data Engineer Intern • Processed 20+ datasets and 10,000+ labeled records with Python/SQL; produced seven reports and deployed MySQL/Hadoop/Spark on a private cloud.	Mar. – Sep. 2024
Zhongzheng Tiantong CPA Tax and Business Consulting Intern Analyzed financial and risk data with Python/SQL and Tableau.	Jul. – Sep. 2023
Feitian Technologies Product Data Engineer Intern Maintained product data and supported C/C++ feature testing and performance reports.	Jul. 2022 – Jan. 2023

Quantitative and Machine Learning Research

- **ReLaMix** (co-first author; arXiv 2026): developed residual mixing for delayed financial observations and evaluated second-level PAXGUSDt forecasts, BTCUSDt transfer, and delay robustness with non-overlapping temporal splits.
- **Wind energy forecasting** (third author; *Energies* 2025): trained modular echo state networks with clustering; best R^2 0.9905.
- **AI for critical heat flux in energy systems** (review in preparation; planned submission to *Energy*): synthesizing prediction, sensing, validation, uncertainty, and engineering-use criteria.
- **Learning systems and LLM evaluation**: led a federated-learning survey (*Expert Systems* 2026). First-authored a study on reasoning after unlearning submitted to ICLR 2027; co-authored heterogeneous federated-learning, model-evaluation, and multimodal-unlearning submissions to AAAI/ICLR 2027.
- **Industrial defect detection** (co-first author; ICASSP 2026): developed frequency and spatial modules and evaluated cross-domain performance; reported mAP@0.5 improvement of 4.8%.

Selected Publications and Honors

ReLaMix* (arXiv 2026); wind forecasting (*Energies* 2025); FREDNet* (ICASSP 2026); federated-learning survey (*Expert Systems* 2026); TRACE* (ACM MM 2026, accepted). Full list: laitty.github.io/publications/.

* Tianyou Lai is a co-first author.

Lanzhou University academic, internship, and exchange scholarships; honors thesis (2025). English: TOEFL 103.